

SECOND SEMESTER B.A./B.Sc. DEGREE EXAMINATION, APRIL 2020

(CBCSS—UG)

Mathematics

MTS 2B 02—CALCULUS OF SINGLE VARIABLE—I

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer any number of questions.**Each question carries 2 marks.**Maximum 25 marks.*

1. Let f and g be functions defined by $f(x) = x + 1$ and $g(x) = \sqrt{x}$. Find the functions $g \circ f$ and $f \circ g$. What is the domain of $g \circ f$?
2. Find $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$.
3. Let $f(x) = \begin{cases} x^2 - x - 2 & \text{if } x \neq 2 \\ 1 & \text{if } x = 2. \end{cases}$ Show that f has a removable discontinuity at 2. Redefine f at 2 so that it is continuous everywhere.
4. Find $\lim_{x \rightarrow \pi/4} \frac{\sin x}{x}$.
5. Show that $f(x) = |x|$ is continuous everywhere.
6. Find the derivative of $\sqrt[3]{x} + \frac{1}{\sqrt{x}}$.
7. Find the critical points of $f(x) = x^3 - 6x + 2$.
8. Find $\lim_{x \rightarrow \infty} (2x^3 - x^2 + 1)$ and $\lim_{x \rightarrow -\infty} 2x^3 - x^2 + 1$.
9. Find the interval on which $f(x) = x^2 - 2x$ is increasing or decreasing.
10. Find the vertical asymptote of the graph of $f(x) = \frac{1}{x-1}$.

Turn over

11. Find $\int \frac{\cos x}{1 - \cos^2 x} dx$.
12. Find $\int x e^{-x^2} dx$.
13. Suppose $\int_1^6 f(x) dx = 8$ and $\int_4^6 f(x) dx = 5$, what is $\int_1^4 f(x) dx$.
14. Find the volume of the solid obtained by revolving the region under the graph of $y = \sqrt{x}$ on $[0, 2]$ about the x -axis.
15. Find the work done in lifting a 2.4 kg. package 0.8 m. off the ground (given $g = 9.8 \text{ m./sec.}^2$).

Section B

Answer any number of questions.

Each question carries 5 marks.

Maximum 35 marks.

16. Find the slope and an equation of the tangent line to the graph of the equation $y = -x^2 + 4x$ at the point $P(2, 4)$.
17. Suppose that $g(x) = (x^2 + 1)f(x)$ and it is known that $f(2) = 3$ and $f'(2) = -1$. Evaluate $g'(2)$.
18. (a) Show that $f(x) = x^3$ satisfies the hypothesis of the mean value theorem on $[-1, 1]$.
 (b) Find the numbers c in $(-1, 1)$ that satisfies the equation as guaranteed by the mean value theorem.
19. Find the slant asymptotes of the graph of $f(x) = \frac{2x^2 - 3}{x - 2}$.
20. A car moves along a straight road with velocity function $v(t) = t^2 + t - 6$, $0 \leq t \leq 10$, where $v(t)$ is measured in feet per second.
 (a) Find the displacement of the car between $t = 1$ and $t = 4$.
 (b) Find the distance covered by the car during this period.
21. (a) Evaluate $\int_{-3}^0 (x^2 - 4x + 7) dx$ by Fundamental theorem of Calculus.
 (b) Use the definition of definite integral to show that if $f(x) = c$, a constant function, then $\int_a^b f(x) dx = c(b - a)$.
22. Find the center of mass of a system comprising three particles with masses 2, 3 and 5 slugs, located at the points $(-2, 2)$, $(4, 6)$ and $(2, -3)$ respectively.
23. Find the length of the graph of $x = \frac{1}{3}y^3 + \frac{1}{4y}$ from $P\left(\frac{7}{12}, 1\right)$ to $G\left(\frac{67}{24}, 2\right)$.

Section C

*Answer any two questions.
Each question carries 10 marks.*

24. (a) Find $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta}$.
- (b) Use intermediate value theorem to find the value of c such that $f(c) = 7$, where $f(x) = x^2 - x + 1$ on $[-1, 4]$.
- (c) In a fire works display, a shell is launched vertically upwards from the ground, reaching a height $S = -16t^2 + 256t$ feet after t seconds. The shell burst when it reaches its maximum height :
- (i) A what time after launch will the shell burst.
- (ii) What will be the altitude of the shell when it explodes ?
25. Find the dimensions of the rectangle of greatest area that has its base on the x -axis and is inscribed in the parabola $y = 9 - x^2$.
26. Using the definition of definite integral evaluate $\int_a^b x dx$.
27. Find the area of the surface obtained by revolving the graph of $f(x) = \sqrt{x}$ on the interval $[0, 2]$ about the x -axis.

$(2 \times 10 = 20 \text{ marks})$